

Postdoctoral Fellowship Application

NSF Atmospheric and Geospace Sciences Postdoctoral Research Fellowship

Department of Geography, University of Florida

Mostafa Rezaali

Ph.D. Candidate, Climate Science (Geography)

University of Florida

mostafarezaali@gmail.com | (352) 709-5350

Budget Justification

Overview

This AGS-PRF fellowship budget follows the solicitation-specified fellowship structure. The budget contains stipend support for the Fellow and a fellowship allowance. No voluntary committed cost sharing is included.

Year 1

Stipend: \$70,000. The stipend will support full-time fellowship effort by Mostafa Rezaali on the research, broader-impact, mentoring, dissemination, and professional-development activities described in the Project Description. Activities include developing the conditional-flow forecast model, preparing the cross-validated hindcast archive, computing deterministic and probabilistic verification metrics, and preparing the first manuscript.

Fellowship allowance: \$30,000. Planned uses include health insurance and benefit-related costs; high-performance computing, cloud storage, and secure data-storage expenses; workstation or peripheral support needed for model development and analysis; conference and workshop travel to present results and receive feedback from the atmospheric-science and GeoAI communities; open-access publication and data-archive fees; and modest software, training, and dissemination costs directly related to the fellowship.

Year 2

Stipend: \$72,000. The stipend will support full-time fellowship effort during the second year, including probabilistic ensemble training, uncertainty-calibration analyses, physical-regime diagnostics, manuscripts, open-science release, student mentoring, and career-development activities.

Fellowship allowance: \$30,000. Planned uses mirror Year 1 and will support continued health/benefit costs, computing and storage, conference participation, publication and data dissemination, and project-specific supplies or software. The allowance will be administered responsibly by the Fellow in accordance with NSF and AGS-PRF requirements.

Budget Rationale

The requested resources are necessary because probabilistic graph-based climate forecasting requires substantial computing, data storage, reproducible workflow development, and dissemination. The fellowship allowance will enable training and evaluation of ensembles, preservation of derived diagnostics, presentation of results to the atmospheric-science community, and open release of reusable scientific products. All requested items are directly aligned with the research and broader-impact activities of the fellowship.